

Comparison of Physiological Cues for Cognitive Load Measures in VR

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ABSTRACT

Cognitive load (CL) is an important measure for detecting whether a user is struggling with a task. Many researchers have investigated physiological cues for measuring CL, such as pupil dilation, galvanic skin response (GSR), heart rate, and electroencephalography (EEG). However, physical activity can often affect the measured signals, so it is unclear how suitable these measures are for use in Virtual Reality (VR). In this paper we present a VR game, PlayMeBack, for measuring users' CL using different physiological cues. In the game, the user is shown different patterns of tiles lighting up and is asked to replay the patterns by pressing tiles in the same sequence. Task difficulty increases with the length of the lighting pattern. We found that for this type of VR environment, EEG and GSR are the most reliable physiological cues for measuring the user's cognitive load, whereas pupil dilation and heart rate did not vary significantly with increasing task difficulty.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality

1 INTRODUCTION

A person's cognitive load (CL) often increases while performing a mentally challenging task. Previous research has shown that CL variation can be measured by physiological cues, such as pupil dilation [3], Heart Rate (HR) [4], galvanic skin response (GSR) [5] and Electroencephalogram (EEG) [1]. However, it is unclear how suitable these measures are for use in Virtual Reality (VR).

In this paper we investigate, how can we use physiological cues to measure cognitive load within a VR environment, and which physiological cues are most reliable for this? This is important because there is limited research on the measurement of cognitive load in VR environments using physiological cues. Previous research studying cognitive load within VR suffers from several limitations. Many previous VR studies used tasks requiring a long duration (several minutes), whereas psychology studies investigating CL without VR typically use stimuli lasting only 2 to 5 seconds. Since users are physically more active in VR tasks, user muscle movements or eye blinks in VR studies can interfere with EEG output and CL measures. So one of the novelties of this study is our special setup requiring only moderate movement and response times, which eliminates these effects and minimizes muscle movement rejection, especially in EEG data analysis. With these limitations, there is a need to investigate

how reliable physiological cues are in an interactive immersive VR application.

2 DESIGN AND IMPLEMENTATION

We designed a VR game called "PlayMeBack" and used it for a user study consisting of a study phase and an encoding phase separated by a 3 second rest. In the study phase, the user in VR sees different patterns of lit tiles (in random sequences of length 3 to 6). In the encoding phase, the user is asked to play the patterns back in the same order as they were lit. The number of tiles that the user has to remember in each task establishes the task difficulty, increasing the cognitive load as the number of tiles increases. The player can freely choose what tiles to playback and each task normally takes 2-5 seconds to complete. The study consisted of 80 trials, 20 for each of the patterns with lengths 3-6.

For our system, we used the HP Omnicept¹ head mounted display (HMD) to display a VR environment developed using Unity3D. The Omnicept HMD has integrated eye-tracking and software for measuring pupil dilation. The Leap Motion² controller is mounted on the headset to track the player's hand interacting with the tiles. EEG data is collected using an Enobio 20 gel cap³ (on 10/20 standard location). GSR and heart rate data was collected using a Shimmer⁴ sensor worn on the wrist.

The EEG, HR and GSR data were sent via the Lab Streaming Layer (LSL) to Unity3D to add physiological markers. The pupil dilation was collected through the HP Omnicept SDK and added with physiological markers. Each marker includes the beginning and information about the task that players were supposed to play. The data was then stored in csv format for further analysis.

3 RESULT

In this study we had 19 participants, 13 male, 6 female, mean age 23.5 years old. The user study took 80 minutes (including EEG cap preparation and setting). Participants were given an 20\$ super-market voucher as compensation. The user study was approved by the University of Auckland Human Participants Ethics Committee, Reference number UAHPEC23449.

To process the recorded data, we first categorized the responses into 3/4/5/6 difficulty level tasks, and divided them into "Hit" or "Miss". A Hit is when a player hits the tiles in the correct order, otherwise it is a Miss. For this study we just focused on "Hit" responses.

Our performance analysis shows that the success rate (Hit) decreases with increasing difficulty: 97%, 95%, 90%, and 79% hit rate

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¹<https://www.hp.com/us-en/vr/reverb-g2-vr-headset-omnicept-edition.html>

²<https://www.ultraleap.com/product/leap-motion-controller/>

³<https://www.neuroelectrics.com/solutions/enobio/20>

⁴<https://shimmersensing.com/product/consensus-gsr-development-kits/>

for 3/4/5 and 6 combination patterns, respectively. This confirms that differences in workload are indeed experienced: with an increase in the number of combinations in a pattern (difficulty level), task performance decreases.

To analyse GSR data, we created bins using markers (Hit/Miss) for each 3/4/5/6 difficulty level and calculated a baseline for each task. The delta value was calculated by subtracting the averaged skin conductivity (SC) from its baseline value. The average delta SC was calculated by averaging GSR over 17 participants in each difficulty level (Two participant's GSR data didn't record due to disconnection). The result shows an increasing SC trend from a 3 combination to a 6 combination pattern (with average delta SC values of -0.018, -0.001, 0.003 and 0.054 accordingly). Our two-pair ANOVA results show a significant difference in all difficulty levels, comparing the 3 vs. 6 combination pattern ($p = 0.009$), comparing the 4 vs. 6 combination pattern ($p = 0.017$) as well as comparing the 5 vs. 6 combination pattern ($p = 0.026$).

To analyse the pupil dilation data, we calculated the baseline value by averaging 500 ms of dilation data at the end of the 3 second rest period then the pupil dilation delta was calculated by subtracting the average pupil dilation value from its baseline value for each eye. The average delta value was calculated over 19 participants. The mean of the 3 combination pattern for the right and left pupil dilation was -0.19 (SD = 0.11) and -0.21 (SD = 0.11). The mean of the 4 combination pattern for the right and left pupil dilation was -0.17 (SD = 0.12) and -0.19 (SD = 0.12); The mean of the 5 combination pattern for right and left pupil dilation was -0.17 (SD = 0.12) and -0.18 (SD = 0.12). The mean of 6 combination pattern for right and left pupil dilation was -0.18 (SD = 0.13) and -0.20 (SD = 0.13). Our two-pair ANOVA result shows no significant difference in any difficulty conditions.

Due to HRV analysis limitations, to analyse the HR data, we just used an averaging method for our HR data. Data processing was done by calculating the grand average for each difficulty level over 12 subjects (due to massive sensor disconnection, we end up with only 12 subjects who had reliable data in all four conditions). Using an ANOVA pair analysis, we found no significant difference in any conditions.

To analyse the EEG data, we used the EEGLAB Toolbox [2]. The pre-processing steps started with categorizing the responses into different difficulty levels (3/4/5/6) based on the length of a lit tile sequence. For each task we created "Hit" and "Miss" bins. We pre-processed the EEG data using markers as follows: (1) Re-referencing EEG data to average mastoids; (2) High-pass and low-pass filtered at 1 Hz and 50 Hz; (3) Down sampling data to 256 Hz; (4) Using the Independent Component Analysis (ICA) to identify and eliminate the blink noise; (5) Create bins using the "Binlister" for each difficulty level to separate and create a total of 80 epochs, 20 epochs from each difficulty level (each epoch contained 2 to 3 seconds of EEG data, identified from the markers); (6) Rejecting epochs affected with saccade and muscle movements which varied for each subject (a rejection rate of 15%, using a "sample-to-sample voltage threshold" and the "simple voltage threshold" artifact rejection function tool).

The post-processing steps are as follows: (1) Create a study using "study" function from EEGLAB and add all pre-process EEG data from all of the participants; (2) Pre-setting the design function for our four CL difficulty conditions using labels, set the epochs baseline to -50ms to 0ms of epochs and set FFT to compute the Power Spectrum for our channels. To produce the result, after computing the grand average for all participants at each level of difficulty, we used the EEGLAB function to measure and run our statistical analysis.

To determine whether we can measure the CL (difficulty) based on the pattern length of a task, we used band frequency analysis. Figure 1, illustrates the power spectral density of different bands at the F4 location (N = 17). For example, the black box shows the Alpha band on comparing 3/4/5 and 6 combination patterns

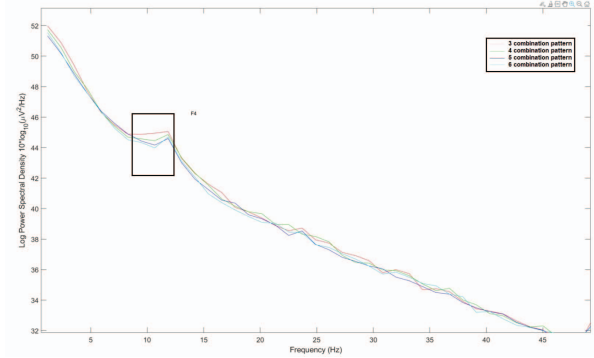


Figure 1: The power spectral density (PSD) at difference frequencies comparing the 3/4/5 and 6 combination patterns (difficulty levels) at the F4 location, the black box shows the Alpha band.

(difficulty levels).

4 DISCUSSION AND CONCLUSION

The main novelty of this study was (1) to use tasks with different difficulty levels, short response times, and requiring moderate muscle movements that closer resemble real learning tasks and (2) to show how reliable and accurate it is to measure and analyse the CL from these difficulty levels by using different physiological cues in VR. We used a different approach to analyse our EEG data; PSD was generated to show the difference of PSD for the lower alpha band (8-10 Hz) and upper alpha band (10-13 Hz) in our different conditions which is first decreasing in lower alpha and then increasing in the upper alpha band. That was repeated for all of the difficulty levels, so we conclude that we do not need to use an absolute or mean alpha for Alpha EEG band analysis. Our main motivation was to show this trend of PSD that appeared in the most frontal and central regions is clearer in electrodes F4, F8, C4 and T8.

Overall we observed the highest PSD of the alpha band for the lowest CL in the 3 combination pattern and the lowest PSD of the alpha band for the highest CL in the 6 combination pattern. This significant alpha band effect is more clear and significant ($p < 0.05$) at 11Hz frequency in the frontal and central regions but it wasn't significant in any other alpha band frequencies. Based on this result we suggest that power in the alpha band could be used as a reliable indicator of CL levels and EEG could be one of the best physiological cues for CL measurement. Our results also show that SC is the second-best physiological cue for the measurement of CL. However, these results are specific to this tile pattern task, so in the future we will need to explore other CL tasks and a wider range of physiological cues.

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